Formulas

InferMesh gateway. Defaults unless named. *Assumed* means used and unpublished. Simulator-only identities at the end.

Notation. Pool W for a model: breaker admits the worker and waiters $< Q$. Load $L(w) =$ requests in flight. Prompt string p. Match depth d. Attempts $A = 2$.

1. Prefix

$$p = \text{join}(\texttt{role} : \texttt{content})$$

over `messages`, newline-separated (`RadixTrie.extract_prefix`). Completions use the raw `prompt` string. Embeddings skip the trie.

2. Token reserve (ETIF)
Heuristic, not a tokenizer. Let c be the sum of `content` lengths in `messages` (roles are not counted).

$$\hat{n} = \max\Big(1, \Big\lfloor \frac{c}{4} \Big\rfloor\Big) + \texttt{max_tokens}$$

(or 256 if the client omits `max_tokens`). Set `etif_count_output_tokens=false` to drop the second term. Reserved until the response ends. c is not $|p|$: the trie string includes `role:` .

3. Selection
Problem. Send p to a worker that already holds a long prefix of p, unless that pins load.

SGLang cache-aware (`sgl-model-gateway`). CLI defaults: cache-threshold 0.3, balance-abs-threshold 64, balance-rel-threshold 1.5. Load is in-flight count. Imbalanced iff both

$$\max L - \min L > 64 \quad \text{and} \quad \max L > 1.5 \min L.$$

Otherwise a hit iff there exist live tenants of the longest match with

$$d \geq \texttt{kv_match_min_chars} \quad \text{and} \quad d > 0.3\ |p|.$$

Then pick $\arg\min L$ among those tenants.

Here. Same predicates. On miss or imbalance: power-of-two choices, not SGLang's min-load (random tie-break). Match rate in SGLang is $d/|p|$ in characters (`prefix_match_with_counts`). Same here.

Insert of p on dispatch (`insert_on_dispatch=true`), as SGLang inserts at `select_worker`.

4. Power of two choices
Theorem (Azar–Broder–Karlin–Upfal, SIAM J. Comput. 29 (1999)). n balls, n bins, each ball into the least loaded of $d \geq 2$ uniform bins: max load $(\log\log n)/(\log d) + O(1)$ w.h.p. For $d = 1$ it is $\Theta(\log n / \log\log n)$.

Supermarket (Mitzenmacher, IEEE TPDS 12 (2001)). Poisson arrivals rate λn, $\lambda < 1$, exponential service; $d = 2$ cuts expected wait exponentially vs $d = 1$.

Here. $d = 2$. Sample without replacement, weights = worker `weight`. Compare scores, not raw queue length. The theorem applies only while the score is a live queue; it does not cover a stale or mixed score.

5. Score
Let $T(w)$ be tokens in flight, $U(w) \in [0, 1]$ engine KV util, $q(w)$ engine wait depth.

$$s(w) = \begin{cases} \min(T(w)/S, 1) & \text{if openai or } \ U \ \text{ stale} \\ \alpha \min(T/S, 1) + \beta \min(q/q_0, 1) + (1 - \alpha - \beta)U & \text{otherwise} \end{cases}$$

Defaults: $\alpha = 0.6$, $\beta = 0$, $q_0 = 8$, $S = 20 \cdot 1024$ (or `etif_scale`). Stale means older than 3 poll intervals. $\alpha = 0.6$ is *assumed* (no published mix). With $\beta = 0$ the vLLM term is $0.4U$.

6. Trie
Character radix (Morrison PATRICIA, 1968): one child per first character, split on LCP. Tenants stamped on every node of an insert. Match may stop inside an edge.

TTL 300 s per attribution. Cap 2^{26} characters per worker (SGLang `max_tree_size`). Over cap: drop LRU leaves until 0.9 of the cap (*assumed* headroom; SGLang uses a periodic node LRU).

LCP length: binary search on slices, $O(\log n)$ Python steps, $O(n)$ bytes compared.

7. Retry budget
SRE (*Site Reliability Engineering*, Ch. 21): retry while the retry ratio stays below 10 percent; amplification at most about 1.1 vs about 3 with a per-request cap of three.

Envoy `RetryBudget`: default 20 percent of active+pending, floor `min_retry_concurrency=3`.

Here. Credit ledger, mode `static`:

$$c \leftarrow \min(c_{\max}, c + 0.10) \quad \text{per client request}$$

$$\rho = \rho_0\big(1 - \alpha_\rho h\big), \quad \rho_0 = 1,\ \alpha_\rho = 0.$$

Admit a retry iff $c \geq \rho$. Then $c \leftarrow c - \rho$. With $\alpha_\rho = 0$, $\rho = 1$: one retry per ten requests in the mean, burst $c_0 = 3$, cap $c_{\max} = 10$. $h = d_w / \ |p|$ for the fallback worker; unused while $\alpha_\rho = 0$.

Mode `headroom` (*assumed*, off): deposit $\max(0, \min(R/(\rho_0\lambda) - 1, 1))$ once λ is known over a window of at least 1 s. Local reroutes (open breaker, full queue) skip the ledger.

Hard cap $A = 2$ attempts, distinct workers. Retry only before the first byte, on 429, 5xx, or transport. Timeouts are not retried.

8. Circuit breaker
Envoy outlier. `consecutive_5xx=5`. Real ejection time = `base_ejection_time` times consecutive ejections, capped by `max_ejection_time`. Defaults 30 s and 300 s.

$$T(k) = \min(T_0 k, T_{\max}), \quad T_0 = 30,\ T_{\max} = 300.$$

First open $k = 1 \Rightarrow T = 30$ s. Success sets $k = 0$. Explicit `Retry-After` cool-down does not increment k.

States: CLOSED → OPEN after 5 consecutive 5xx (not 429) → HALF_OPEN after T, one probe. 429 opens for the header delay, unmultiplied.

Not taken from Envoy: `max_ejection_percent` (default 10 percent); the interval that decrements the multiplier while healthy (here a success zeros k).

9. Cooldown headers
RFC 9110 §10.2.3. `Retry-After = HTTP-date / delay-seconds`. Invalid ⇒ ignore.

Else the largest vendor reset (`x-ratelimit-reset-*`, Anthropic RFC 3339 / epoch). Cap 300 s. Never ∞.

10. Queue
Semaphore $C = 20$ in-flight per worker. Waiters $\leq Q = 40$, wait ≤ 30 s, else 503 with `Retry-After`. Routable while waiters $< Q$.

11. Replay / retry benches (not the gateway)
Prefill time, DistServe (arXiv 2401.09670) Appendix A *form* $T = C_1(\cdots) + C_2(\cdots) + C_3$, constants fitted per model/GPU and unpublished:

$$t_{\text{prefill}(u)} = c_1 u + c_2 u^2 + c_3$$

with $c_1 = 4 \cdot 10^{-5}$, $c_2 = 3 \cdot 10^{-9}$, $c_3 = 0.02$, ITL 0.02 s *assumed*. Claim policy order, not seconds.

SLO in `bench/replay/metrics.py`: TTFT 0.25 s, TPOT 0.10 s (DistServe Table 1, OPT-13B chatbot). `docs/benchmarks.md` reports a 2 s TTFT cut on the long Mooncake prompts.

Retry sim: Bronson et al., HotOS 2021 §2.1, capacity 300 QPS, offered 280 QPS.

12. Sources

Object	Statement	Here

Cache hit / imbalance	SGLang CLI `router_args.py` $0.3/64/1.5$; `cache_aware.rs`	same predicates; miss $\to$ P2C
$d = 2$ max load	Azar et al. 1999	score, not queue
Retry ratio 10 percent	SRE Ch. 21	deposit 0.10, $\rho = 1$
$T = T_0 k$	Envoy outlier	`proxy.py` `CircuitBreaker._open`
`Retry-After`	RFC 9110 §10.2.3	`ratelimit.py`
Prefill C_i	DistServe App. A	replay only, C_i assumed
$\alpha = 0.6$, α_ρ, headroom 0.9	$-$	*assumed*

Compile: `typst compile docs/formulas.typ docs/formulas.pdf`